

Tinker Tailor LLM Spy: Reconstructing "Deleted" Chats & Hijacking Sessions from Chromium LevelDB Caches

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Abstract

As enterprise adoption of AI coding assistants and browser LLM portals (ChatGPT, Claude AI, Gemini, DeepSeek, Perplexity, Cursor, VS Code Copilot) explodes, local client-side storage has become a primary target for adversary post-exploitation and a major corporate DLP risk. Employees routinely input proprietary source code, credentials, and PII, assuming session deletion removes traces. In this paper, we present Verity, a sub-50ms forensic carving framework that bypasses traditional file-carving limitations to reconstruct volatile LLM artifacts directly from V8 LevelDB SSTables, Claude TipTap draft keystroke buffers, and IDE JSONL transcripts. We reveal how a 41ms headless infostealer PoC (verity_stealer.py) can silently harvest unsubmitted prompts and active session tokens before EDR tools register file modification anomalies. Finally, we evaluate a real-time Shadow AI DLP engine that parses carved chats on-the-fly and alerts on credentials and intellectual property leaks directly inside the incident response view.

I. INTRODUCTION

The rapid integration of Large Language Model (LLM) interfaces into software engineering workflows has introduced a novel threat vector. Web-based portals and IDE extensions process high-value intellectual property, including proprietary source code, internal network architecture, database connection strings, and credential tokens.

User operations rely on client-side storage mechanisms—specifically IndexedDB LevelDB on Chrome/Edge platforms, tipTapEditorState JSON buffers on Claude AI, and JSONL agent transcripts on IDE assistants like Cursor and VS Code Copilot. A critical architectural blind spot exists: deleting a chat session through the user interface updates high-level metadata pointers but leaves underlying raw payload data fully intact in non-volatile LevelDB SSTables and log files.

II. SUB-50MS INFOSTEALER VECTOR & THREAT MODEL

To evaluate the latency risk of client-side LLM storage, we constructed a zero-dependency, headless Python infostealer proof-of-concept (verity_stealer.py). Operating entirely in-memory without invoking heavy browser debugging drivers, the script scans local IndexedDB directories and IDE storage paths.

Empirical testing demonstrates that verity_stealer.py achieves complete extraction of 485+ prompt records, unsubmitted keystroke buffers, and active session cookies in **41 milliseconds (0.041s)**. This sub-50ms latency allows adversaries to harvest sensitive telemetry during transient execution windows before standard Endpoint Detection and Response (EDR) agents trigger anomalous disk I/O alerts.

III. SYSTEM ARCHITECTURE & CARVING MECHANICS

Verity operates via three specialized carving sub-engines designed to parse heterogeneous storage formats:

- **V8 LevelDB Varint Carving:** Scans Chrome/Edge SSTables and log blocks, decoding V8 varint integer lengths and reconstructing full multi-turn conversation trees for ChatGPT, Gemini, DeepSeek, and Perplexity.
- **Claude TipTap Draft Keystroke Recovery:** Scans LevelDB value buffers for tipTapEditorState JSON structures, extracting text typed into the browser prompt box before the user clicks 'Send'.
- **IDE Agent Transcript Carving:** Traverses local project storage paths for Cursor (.cursor/projects/) and VS Code Copilot (globalStorage/*ChatSessions/), parsing raw JSONL prompt histories.

IV. EMPIRICAL BENCHMARK EVALUATION

We evaluated Verity against standard open-source and commercial forensic toolsets (Autopsy, Magnet AXIOM, and Volatility) across 2,880+ carved records.

Capability / Feature	Autopsy / Magnet AXIOM	Verity Forensics Framework
ChatGPT V8 Tree Recovery	Generic Strings Only	100% Tree & Node Recovery

Claude TipTap Keystroke Drafts	0 Records Recognized	Prefix Deduplicated Recovery
Cursor & Copilot JSONL Parsing	Unsupported Schema	Native Agent Crawler
Secret Leak Detection & DLP	Basic YARA Regex	Real-time Scanner + UI Badges
Execution Speed	Minutes to Hours	51.9 milliseconds

V. CONCLUSION & DEFENSIVE COUNTERMEASURES

Verity demonstrates that local AI telemetry persistence presents a significant, unmonitored attack vector in modern enterprise networks. To mitigate these risks, organizations should enforce client-side database encryption-at-rest, mandate explicit LevelDB zeroization upon chat deletion, and deploy endpoint detection rules monitoring non-standard access to browser IndexedDB paths.